IMovies – Movie Ticket Booking System

Ideation Phase

Project concept, motivation and scope

Project documentation based on the uploaded bookMyShow-main source ZIP

Prepared: 31 October 2026

Ideation Phase

1.1 Project overview

IMovies is a web-based movie discovery and ticket-booking application inspired by online cinema-booking platforms. The uploaded code provides user registration and login, movie listings, theatre/show management, seat selection, booking records and administrative management screens.

1.2 Problem statement

Traditional ticket booking can require customers to visit a cinema or contact staff to check movie availability and reserve seats. A centralized web application can make movie discovery, show selection and seat reservation more convenient while helping theatre operators manage listings and shows.

1.3 Objectives

- Provide a browser-based interface for discovering available movies.
- Allow users to register, log in and access protected account features.
- Display theatres and show schedules associated with movies.
- Allow a user to select seats and submit a booking.
- Store booking and show information in MongoDB.
- Provide administrative workflows for managing movies, theatres and shows.

1.4 Proposed solution

A React single-page application communicates with an Express REST API. The API reads and writes MongoDB documents through Mongoose. JWT-based authentication is used for protected routes, and passwords are hashed with bcryptjs during registration.

1.5 Existing vs proposed system

Aspect	Conventional/manual approach	Proposed IMovies application
Discovery	Ask cinema staff or browse separate sources	Movie listings in one interface
Show selection	Manual enquiry	Theatre and show information in the app
Seat booking	Counter-based/manual process	Seat-grid selection and booking API
Records	Paper or disconnected records	MongoDB-backed user and booking records
Administration	Manual updates	Protected movie/theatre/show management screens

1.6 Scope

In scope: user accounts, movie catalogue, theatre information, shows, fixed 10×10 seat layout described in README, selection of up to six seats per booking, booking history, confirmation screen and admin management. The source code should be checked locally to confirm each flow.

1.7 Limitations and out of scope

- The supplied code uses a fixed seat layout rather than configurable auditorium layouts.
- The README describes a maximum of six seats per booking.

- A production-grade payment gateway, refunds, email/SMS notifications, QR ticket scanning and mobile applications are not established as complete features by the source inspection.
- No automated test suite or performance benchmark is included in the server package scripts.

1.8 Intended users

Primary users are moviegoers. Theatre operators or administrators manage theatre, movie and show information.

Accuracy note: This document describes the code and README found in the supplied ZIP. Items marked as recommendations or proposed tests are not claims that they have already been implemented or passed. Confirm all workflows on your laptop before submission.